ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. CSE ENGINEERING — SEMESTER 7

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain CIA Triad and OSI Security Architecture. Describe Passive vs Active attacks. [14 Marks]

[OR]

Q.1 (Alt) Explain Classical Encryption techniques: Caesar cipher, Playfair cipher, Hill cipher, and Vigenere cipher. [14 Marks]

--- UNIT II ---

Q.2 Explain Advanced Encryption Standard (AES) rounds and Data Encryption Standard (DES) Feistel structure. [14 Marks]

[OR]

Q.2 (Alt) Compare Block cipher modes: ECB, CBC, CFB, OFB, and Counter (CTR) modes. [14 Marks]

--- UNIT III ---

Q.3 Explain RSA algorithm with key generation and numerical example. Discuss RSA attack vectors. [14 Marks]

[OR]

Q.3 (Alt) Explain Diffie-Hellman Key Exchange algorithm. How does Man-in-the-Middle attack compromise it? [14 Marks]

--- UNIT IV ---

Q.4 Explain Cryptographic Hash Functions: SHA-256 and MD5 properties. Describe Digital Signature Standard (DSS). [14 Marks]

[OR]

Q.4 (Alt) Describe HMAC and Kerberos authentication protocol version 5. [14 Marks]

--- UNIT V ---

Q.5 Explain Web Security: SQL Injection, Cross-Site Scripting (XSS), and CSRF vulnerabilities with countermeasures. [14 Marks]

[OR]

Q.5 (Alt) Describe Firewalls (Packet filtering, Stateful inspection) and Intrusion Detection Systems (IDS/IPS). [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****